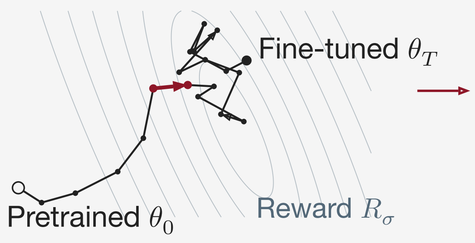
Adaptive and Neutral Theory of Evolution Strategies for Large Language Models

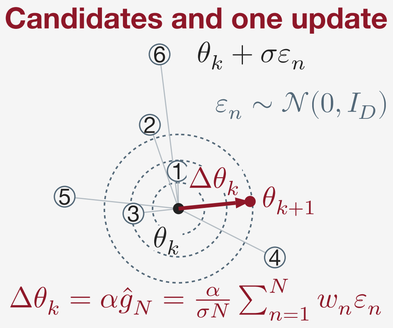
Soma Yokoi · Issei Sato The University of Tokyo

Setup LLM fine-tuning with evolution strategies (ES)

LLM parameter trajectory

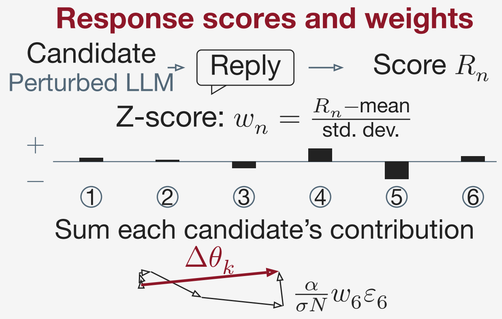


Candidates and one update



$$\Delta\theta_k = \alpha\hat{g}_N = \frac{\alpha}{\sigma N}\sum_{n=1}^{N} w_n\varepsilon_n$$

Response scores and weights



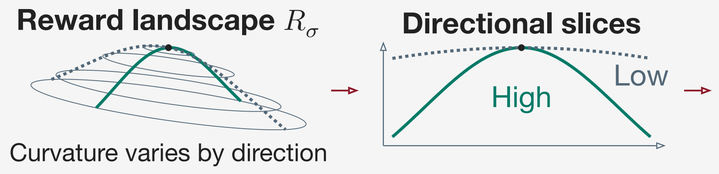
Questions

Why can a **small population learn effectively**?
$D \approx 3 \times 10^9$ parameters; $N = 30$ individuals per update

Why can **performance decline** as training continues?
Performance: the training reward being maximized

Background

LLM reward landscapes: curvature concentrates in a few directions.
Liang et al., 2026



Hypothesis Directions play different roles

Improvement in a few high-curvature directions.
Fluctuations accumulate across many low-curvature directions.

Framework A unified analysis of learning and performance change

1 Stochastic differential equation (SDE)

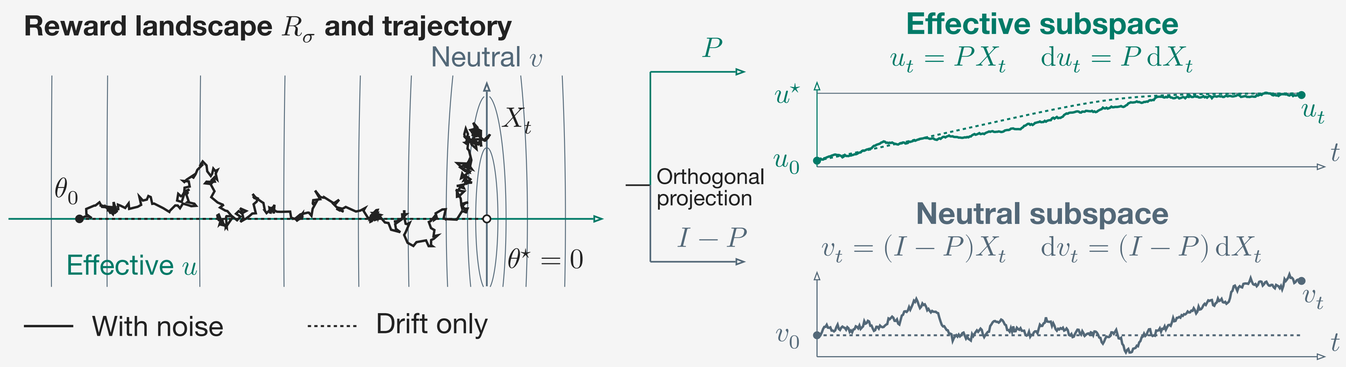
$$\theta_{k+1} = \theta_k + \alpha\hat{g}_N(\theta_k)$$

↓ Continuous-time approximation: $X_0 = \theta_0$

$$\mathrm{d}X_t = \underbrace{\alpha b_N(X_t)\,\mathrm{d}t}_{\text{Drift}} + \underbrace{\alpha A_N(X_t)\,\mathrm{d}W_t}_{\text{Diffusion}}$$

$b_N = \mathbb{E}\hat{g}_N$, $A_N A_N^\top = \mathrm{Cov}(\hat{g}_N)$
t: iteration time; W_t: standard Brownian motion

2 Decompose motion by curvature

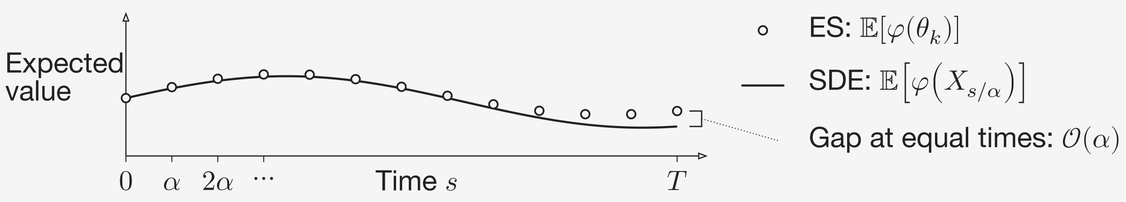


Main results We establish finite-time weak approximation and derive conditions for learning with small populations and subsequent performance decline.

Theorem 1 Finite-time weak approximation of ES

For a test function φ and time horizon T,

$$\max_{0 \le k\alpha \le T} |\mathbb{E}[\varphi(\theta_k)] - \mathbb{E}[\varphi(X_k)]| = \mathcal{O}(\alpha) \quad (\alpha \to 0).$$



Interpretation

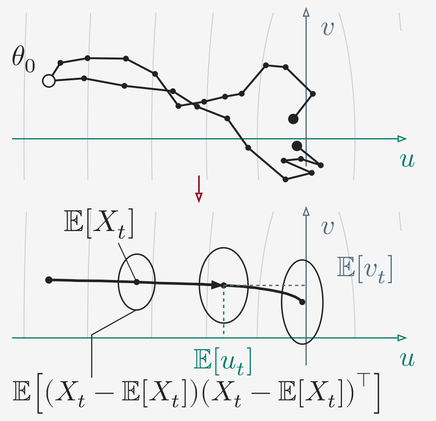
Approximate the mean and covariance of ES with an SDE

↓

Decompose motion by curvature to assess its effect on reward

↓

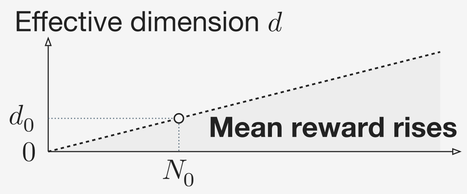
Study small-population learning and later performance decline



Theorem 2 Learning with small populations

Mean reward increases in effective directions if

$$d < \frac{2\sigma N}{L\alpha}\left[\sqrt{2\mu V_u} - \left(\frac{\sigma}{\alpha}\right)\bar{r}\right]$$



$$d = d_0: \quad N > N_0 = \frac{L\alpha d_0}{2\sigma\left[\sqrt{2\mu V_u} - \left(\frac{\sigma}{\alpha}\right)\bar{r}\right]}$$

$V_u = R_\sigma(u^*, 0) - R_\sigma(u, 0)$. Local PL condition (μ) and smoothness (L); $\bar{r}$ bounds the coefficient-approximation correction.

Interpretation

A smaller effective dimension d allows a smaller population N

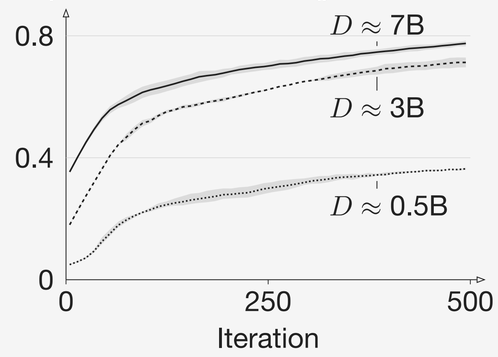
↓

In LLM reward landscapes, d is far smaller than the full dimension D

↓

Small ES populations can optimize very large LLMs

Full-parameter fine-tuning · $N = 30$
Population-mean training reward



Qwen2.5-Instruct · Countdown

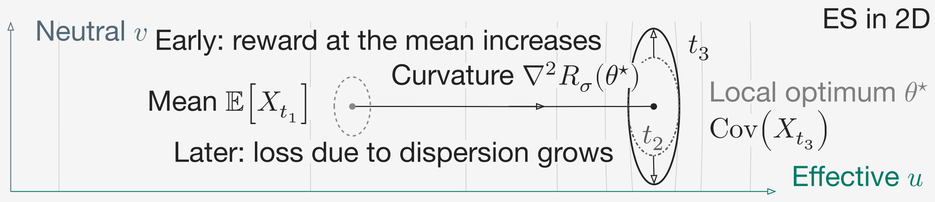
Theorem 3 Reward decline with continued training

Sufficient condition for increasing expected training reward R_σ $(t_1 < t_2)$

$$[R_\sigma(\mathbb{E}[X_t])]_{t_1}^{t_2} > -\tfrac{1}{2}\left[\mathrm{Tr}\{\nabla^2 R_\sigma(\theta^\star)\,\mathrm{Cov}(X_t)\}\right]_{t_1}^{t_2} + \quad \varepsilon$$

Reward change at the mean — Increase in loss due to dispersion — Approximation error bound

ES in 2D



Interpretation

While effective-direction gains slow and neutral spread continues,

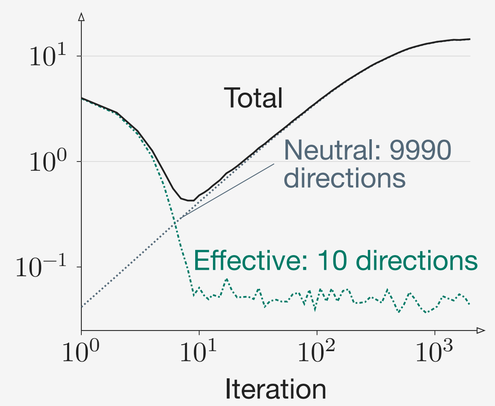
↓

if added neutral-direction losses exceed the remaining reward gain,

↓

expected reward declines with continued training.

Reward gap $\mathbb{E}[R^\star - R(\theta_k)]$



Quadratic $R \cdot D = 10^4$, $d = 10$, $N = 30$

Outlook Toward efficient fine-tuning and continual learning with less forgetting

Notes α: update coefficient; σ: perturbation scale.
Smoothed reward: $R_\sigma(\theta) = \mathbb{E}_\varepsilon[R(\theta + \sigma\varepsilon)]$.